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Model-Contingent Knowledge Extraction from User-Generated Reviews: A Multi-Layer Longitudinal Sentiment Analysis Framework

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Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “Model-Contingent Knowledge Extraction from User-Generated Reviews: A Multi-Layer Longitudinal Sentiment Analysis Framework” for consideration for publication in Data & Knowledge Engineering.

This study proposes a multi-layer longitudinal framework for extracting sentiment-related knowledge from large-scale user-generated reviews in digital marketplaces. Using 250,000 Tokopedia mobile application reviews collected from the Google Play Store between 2018 and 2026, the study investigates how user sentiment evolves over time and how different transformer-based models shape sentiment interpretation.

The manuscript makes several contributions that align with the scope of Data & Knowledge Engineering. First, it conceptualizes sentiment analysis as a data-to-knowledge transformation process, in which unstructured textual reviews are converted into structured and interpretable knowledge about user perception dynamics. Second, it introduces a model-aware analytical framework that integrates temporal analysis, transformer-based sentiment representation using IndoBERT and IndoRoBERTa, model agreement analysis, and lexical pattern extraction. Third, it demonstrates that extracted sentiment knowledge is model-contingent, showing that different transformer architectures may produce systematically different interpretations of the same user-generated content. Finally, the study provides a reusable framework for supporting knowledge extraction, decision support, and continuous monitoring of digital platform performance.

We believe that this manuscript is suitable for Data & Knowledge Engineering because it addresses the transformation of large-scale unstructured data into interpretable knowledge representations, with implications for knowledge extraction, text mining, model-aware analytics, and decision support systems. The proposed framework contributes to the interface between data engineering and knowledge engineering by demonstrating how user-generated textual data can be systematically processed, represented, and interpreted through a multi-layer analytical pipeline.

This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have read and approved the final version of the manuscript and agree with its submission to Data & Knowledge Engineering. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

During the preparation of this manuscript, the authors used ChatGPT and Jenni AI for language refinement, drafting assistance, and editing support. The authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

Thank you for considering our submission. We look forward to your response.

Sincerely,

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Highlights

- A model-aware framework extracts knowledge from user-generated reviews.
- Sentiment dynamics reveal cumulative shifts in user perception.
- IndoBERT and IndoRoBERTa produce model-contingent interpretations.
- Agreement analysis assesses stability of extracted sentiment knowledge.
- Lexical patterns identify drivers of satisfaction and dissatisfaction.

Model-Contingent Knowledge Extraction from User-Generated Reviews: A Multi-Layer Longitudinal Sentiment Analysis Framework

Abstract

This study proposes a multi-layer longitudinal framework for extracting sentiment-related knowledge from large-scale user-generated reviews in digital marketplaces. Using 250,000 reviews of the Tokopedia mobile application collected from the Google Play Store between 2018 and 2026, this study examines how user sentiment evolves over time and how different transformer-based models shape sentiment interpretation. Unlike conventional sentiment analysis approaches that treat sentiment classification as a static predictive task, this study conceptualizes sentiment as a dynamic and model-contingent knowledge representation derived from cumulative user experience. The proposed framework integrates four analytical layers: temporal analysis of rating distributions, transformer-based sentiment classification using IndoBERT and IndoRoBERTa, model agreement analysis, and lexical pattern extraction through bigram and trigram analysis. The findings reveal a structural shift from predominantly positive to increasingly negative sentiment, suggesting that user dissatisfaction emerges progressively as a cumulative outcome of repeated platform interactions rather than isolated service failures. The comparative analysis further shows systematic differences between IndoBERT and IndoRoBERTa, indicating that sentiment knowledge is not methodologically neutral but shaped by model-specific interpretive behavior. Agreement analysis identifies areas of interpretive consistency and divergence, while lexical patterns reveal key drivers of satisfaction and dissatisfaction, including platform usefulness, promotional consistency, system reliability, and expectation–performance alignment. By synthesizing these layers, this study contributes a data-driven framework for transforming large-scale textual reviews into interpretable knowledge about user perception dynamics. The findings highlight the importance of model-aware sentiment analysis as a reusable data-to-knowledge transformation mechanism for knowledge extraction, decision support, and continuous monitoring of digital platform performance.

Keywords: Knowledge Extraction; User-Generated Reviews; Sentiment Analysis; Transformer-Based Models; Longitudinal Analysis; Text Mining

1. Introduction

The rapid expansion of digital marketplaces has fundamentally reshaped how consumers evaluate products and services, particularly within online marketplaces where user-generated content, such as reviews and ratings, increasingly influences decision-making processes (Roma & Aloini, 2018). In digital ecosystems, consumers actively create and disseminate evaluative information, transforming platforms into dynamic spaces of continuous feedback exchange (Kim et al., 2021; Mariani & Nambisan, 2021). This shift has elevated user-generated content from a supplementary information source to a central mechanism influencing trust formation, perceived value, and purchasing decisions (Jin & Phua, 2015; Roma & Aloini, 2018). Prior studies

highlight that textual reviews provide richer and more nuanced insights than numerical ratings, as they capture context-dependent experiences and multidimensional perceptions that cannot be reduced to simple quantitative measures (Hasan et al., 2025). Consequently, understanding user sentiment has become a strategic necessity for digital platform sustainability and competitiveness (Nwaimo et al., 2024).

The Indonesian e-commerce landscape provides a particularly compelling context for examining these dynamics due to its rapid growth (Ahdiat, 2026), large user base (Ahdiat, 2023), and intense competition among platforms such as Tokopedia, Shopee, and Bukalapak (Pusparisa, 2020). Within this ecosystem, Tokopedia represents a significant empirical case due to its longitudinal presence and evolving competitive trajectory (Databoks, 2017; Jayani, 2019). Since 2018, Tokopedia has consistently been among the leading e-commerce platforms in Indonesia, supported by strong user engagement, large transaction volumes, and substantial investments (Jayani, 2019; Tanuwijaya & Andrew, 2019). However, the platform's performance has not remained static. In recent years, particularly since 2023, Tokopedia has experienced fluctuations in user engagement and competitive pressure from rival platforms, reflecting shifts in promotional strategies (Ahdiat, 2023, 2025). This longitudinal trajectory—from early dominance to increasing competitive pressure—positions Tokopedia as a critical empirical case for analyzing how user sentiment evolves over time. The platform's scale, diversity of user interactions, and temporal variation in performance make it a data-rich and strategically relevant setting for investigating sentiment dynamics in digital marketplaces.

In response to the growing importance of user-generated content, sentiment analysis has emerged as a dominant methodological approach for extracting meaningful insights from large-scale textual data (Abu-Salih et al., 2020; Ortiz et al., 2019). As a key area within natural language processing (NLP), sentiment analysis enables the classification of subjective expressions into structured categories such as positive, negative, and neutral, facilitating the interpretation of complex and unstructured user feedback (Ligthart et al., 2021). Numerous studies have applied sentiment analysis in the context of Tokopedia (Meifitrah et al., 2020; Warsito & Prahutama, 2020) and other online marketplaces using various machine learning techniques. Early research examined service quality dimensions of Tokopedia using sentiment analysis, identifying key aspects such as reliability and personalization that influence user perception (Sari et al., 2018). Subsequent studies analyzed Tokopedia application reviews to support service improvement and recommendation systems using classification methods such as Naive Bayes (Meifitrah et al., 2020). Other research focused on evaluating consumer opinions toward products and sellers employing machine learning approaches such as Random Forest, highlighting the role of sentiment in assessing product quality and seller performance (Warsito & Prahutama, 2020).

Further developments in the literature extended sentiment analysis to social media contexts, where public opinion toward Tokopedia and other platforms was analyzed using data from Twitter (Fadlil et al., 2024; Kurnianingrum et al., 2025). These studies demonstrated that sentiment expressed in social media provides valuable insights into user perception and platform

performance in real time (Reddy et al., 2025). More recent research has incorporated topic modeling and clustering techniques to identify dominant discussion themes among marketplace users, revealing that consumer discourse varies significantly across platforms and contexts (Widjaja et al., 2023). In addition, methodological advancements have focused on improving classification performance through algorithm optimization, such as Support Vector Machine (SVM) kernel tuning (Fadlil et al., 2024). Comparative studies have also emerged, analyzing sentiment differences between platforms such as Tokopedia and Shopee, demonstrating that variations in sentiment reflect differences in platform performance, user experience, and competitive positioning (Kurnianingrum et al., 2025).

Despite these contributions, several important limitations remain in the existing literature. First, most studies conceptualize sentiment as a static construct, neglecting its temporal evolution and the cumulative nature of user experience (Bastiaansen et al., 2019; Caruelle et al., 2023). Second, prior research typically relies on a single analytical model, implicitly assuming methodological neutrality, whereas model selection may significantly influence sentiment interpretation (Lapesa & Evert, 2014; Le et al., 2024). Third, existing studies often focus on isolated analytical dimensions, such as classification accuracy, topic modeling, or comparative analysis, without integrating these approaches into a unified analytical framework (Camilleri & Miah, 2021; Doogan et al., 2023). As a result, there is still a lack of comprehensive research that simultaneously captures temporal dynamics, methodological variation, and linguistic patterns in user-generated content (Abu-Salih et al., 2020; Tanaltay et al., 2024). From a data and knowledge engineering perspective, this gap reflects the need for analytical frameworks capable of transforming large-scale unstructured textual data into structured, interpretable, and model-aware knowledge representations.

To address these limitations, this study investigates user sentiment dynamics using a large-scale dataset comprising 250,000 user reviews from the Tokopedia application on the Google Play Store, spanning the period 2018–2026. The longitudinal nature of the dataset enables the analysis of structural changes in user perception over time, providing a more comprehensive understanding of sentiment evolution. It employs two transformer-based models, IndoBERT and IndoRoBERTa, to classify sentiment into positive, neutral, and negative categories. These advanced deep learning approaches enable this study to capture contextual and semantic dimensions of user sentiment more effectively than traditional machine learning methods.

Furthermore, this study introduces a multi-layer analytical framework that integrates four complementary components: temporal analysis, comparative model analysis, agreement analysis, and lexical analysis using bigram and trigram extraction techniques. This integrated framework allows a holistic examination of user sentiment by combining behavioral trends, methodological differences, and linguistic patterns within a unified analytical structure. Such an approach provides deeper insights into how user perception evolves, how models interpret sentiment differently, and how specific linguistic expressions reflect underlying user experiences.

Finally, the findings of this study provide important implications for both theory and practice. From a theoretical perspective, this study contributes to the development of sentiment analysis as a multi-dimensional and context-dependent construct. From a practical perspective, the insights derived from user-generated content can support platform providers in identifying critical drivers of user satisfaction and dissatisfaction, improving service quality, and aligning platform performance with evolving user expectations. In an increasingly competitive digital marketplace, the ability to systematically analyze and interpret user sentiment represents a key strategic capability for sustaining long-term competitiveness (Hou et al., 2019).

This study makes several important contributions to the literature on sentiment analysis and digital platform research.

First, it advances the conceptualization of user sentiment as a dynamic, longitudinal construct rather than a static evaluation. While prior research predominantly relies on static, cross-sectional datasets (Biørn-Hansen et al., 2018; Nasrabadi et al., 2024), this study utilizes a large-scale dataset spanning eight years (2018–2026), enabling the identification of structural shifts in user perception over time. This longitudinal perspective provides deeper insight into how user sentiment evolves in response to changing platform conditions and competitive pressures.

Second, this study introduces a multi-layer descriptive analytical framework, integrating temporal analysis, comparative model analysis, agreement analysis, and lexical pattern extraction. Unlike prior studies that focus on a single analytical dimension—such as classification accuracy or topic modeling—this approach provides a holistic understanding of sentiment dynamics, combining behavioral trends, methodological variation, and linguistic structure within a unified framework.

Third, the study contributes methodologically by reframing the role of transformer-based models in sentiment analysis. Rather than treating models as tools for predictive optimization, IndoBERT and IndoRoBERTa are positioned as analytical instruments for interpretive comparison, revealing how different architectures produce systematically different representations of sentiment. This perspective highlights that sentiment analysis is not methodologically neutral, but inherently shaped by model-specific interpretive behavior.

Fourth, this study provides large-scale empirical evidence from an emerging digital economy context, specifically Indonesia's e-commerce ecosystem. Given the limited availability of longitudinal sentiment studies in non-Western markets, these findings offer valuable insights into user perception dynamics in a rapidly evolving and highly competitive digital environment.

Finally, this study identifies key drivers of user satisfaction and dissatisfaction, including system reliability, promotional consistency, and expectation–performance alignment. These findings highlight the importance of continuous monitoring of user-generated content as a strategic resource for sustaining competitiveness in digital marketplaces.

Taken together, these contributions position the study within the field of data and knowledge engineering, where large-scale textual data are transformed into interpretable knowledge representations to support understanding, monitoring, and decision-making in complex digital environments.

2. Data and Methods

2.1 Data Collection and Preprocessing

This study utilizes a large-scale dataset of user-generated reviews collected from the Tokopedia mobile application available on the Google Play Store. The data were retrieved from the official application page, which provides publicly accessible user feedback reflecting real user experiences, perceptions, and interactions with the platform. All data used in this study were publicly available and collected in accordance with platform policies.

Data were collected using Python within the Google Colab environment, employing the google-play-scraper library to automate the extraction process. The scraping pipeline was designed to ensure reproducibility and consistency, capturing review text, rating scores, timestamps, and associated metadata. To maximize data coverage, the scraping process was configured to retrieve up to 500,000 reviews. However, due to platform-imposed constraints and pagination limitations, a total of 250,000 reviews were successfully collected. This dataset represents the maximum retrievable volume under the given technical conditions and is sufficiently large to support robust large-scale analysis.

The dataset spans a longitudinal period from 2018 to 2026, enabling the examination of temporal dynamics in user sentiment. The distribution of reviews across years is inherently uneven, reflecting organic user activity and platform growth patterns. Rather than applying artificial balancing techniques, this study preserves the natural distribution of the data to maintain ecological validity and ensure that observed patterns reflect real-world user behavior.

Prior to analysis, the dataset underwent a structured preprocessing procedure to ensure data quality and analytical reliability. This process included: (1) removal of empty and null entries, (2) exclusion of non-informative reviews such as numeric-only or symbol-only content, (3) filtering for Indonesian-language text, and (4) elimination of duplicate entries. No aggressive normalization (e.g., stemming or lemmatization) was applied, as transformer-based models inherently capture contextual linguistic variations through subword tokenization mechanisms. This approach preserves the authenticity of user-generated language, which is often characterized by informal expressions, abbreviations, and non-standard grammar.

2.2 Sentiment Classification Approach

To analyze user sentiment, this study employs transformer-based natural language processing models, specifically IndoBERT and IndoRoBERTa, pre-trained language models optimized for Indonesian text. These models have demonstrated strong performance in capturing contextual and semantic relationships in informal and user-generated content (Lin & Nuha, 2023; Wicaksono & Rojali, 2026).

This study does not aim to optimize or benchmark model performance. Instead, it utilizes state-of-the-art transformer models as analytical lenses for examining large-scale sentiment dynamics and uncovering patterns in user-generated content. Accordingly, the study focuses on interpretive consistency and pattern identification rather than predictive performance evaluation.

The selection of IndoBERT and IndoRoBERTa is motivated by their complementary architectural characteristics. IndoBERT is based on the Bidirectional Encoder Representations from Transformers (BERT) framework (Lin & Nuha, 2023), known for capturing contextual dependencies through bidirectional encoding (Lin & Nuha, 2023). IndoRoBERTa is a robustly optimized variant of BERT that emphasizes improved training strategies (Anggrainingsih et al., 2024). The use of these two models enables a comparative perspective on how different transformer architectures interpret sentiment.

Sentiment classification was performed using the Hugging Face Transformers library within a Python-based environment. A pipeline-based inference approach processed the dataset iteratively, ensuring scalability and computational efficiency. The models classified each review into one of three sentiment categories: positive, neutral, or negative. In addition to sentiment labels, the models also generated confidence scores representing prediction probabilities.

To ensure consistency across models, label normalization was performed on output labels from IndoBERT (e.g., LABEL_0, LABEL_1, LABEL_2) and then mapped to standardized sentiment categories, while IndoRoBERTa outputs were converted into a unified format. The results from both models were then integrated into a consolidated dataset, enabling direct comparison and downstream analysis.

Given the absence of a manually labeled ground truth dataset, this study adopts a comparative interpretive approach rather than a predictive validation framework. While this limits the ability to directly evaluate classification accuracy, it does not constrain the study's objective of uncovering structural sentiment patterns. Prior studies have demonstrated that IndoBERT and IndoRoBERTa achieve strong performance in Indonesian sentiment classification tasks (Wicaksono & Rojali, 2026). Therefore, the focus of this study is placed on comparative and interpretive consistency rather than predictive accuracy, allowing the analysis to emphasize structural patterns and model-contingent interpretations of user sentiment. Agreement analysis between models is used as an indicator of interpretive consistency, enabling the identification of both convergent and divergent classification patterns. This approach is particularly suitable for large-scale exploratory studies, where the objective is to uncover structural patterns rather than optimize classification accuracy (Bouncken et al., 2021; Bzdok et al., 2019).

227

228 **2.3 Analytical Framework**

229 This study adopts a multi-layer analytical framework designed to capture the complexity of user
230 sentiment in a longitudinal and data-driven context. The framework integrates four
231 complementary analytical components, enabling a comprehensive examination of sentiment
232 dynamics from multiple perspectives.

233 First, a temporal analysis is conducted to examine the evolution of user ratings and sentiment
234 distributions over time. By analyzing year-by-year variations, this component identifies
235 structural shifts in user perception and highlights long-term trends rather than short-term
236 fluctuations.

237 Second, a comparative model analysis is performed to evaluate differences in sentiment
238 classification outcomes between IndoBERT and IndoRoBERTa. This analysis provides insight
239 into how model architecture influences sentiment interpretation, revealing systematic variation
240 in classification behavior.

241 Third, an agreement analysis is conducted to measure the level of consistency between the two
242 models. This component quantifies the proportion of reviews that receive identical
243 classifications, serving as an indicator of interpretive robustness. Patterns of disagreement are
244 also examined to identify areas where sentiment interpretation is more ambiguous or model-
245 dependent.

246 Fourth, a lexical analysis is implemented using bigram and trigram extraction techniques. This
247 approach identifies frequently occurring word combinations in both positive and negative
248 reviews, enabling the exploration of underlying drivers of user satisfaction and dissatisfaction.
249 Unlike purely statistical analysis, this component provides interpretive depth by linking
250 sentiment patterns to specific linguistic expressions.

251 By integrating these analytical layers, the proposed framework moves beyond single-method
252 analysis and provides a holistic understanding of sentiment dynamics. The combination of
253 temporal trends, model comparison, agreement evaluation, and linguistic pattern extraction
254 enables a richer and more nuanced interpretation of large-scale user-generated data. To ensure
255 methodological transparency, the overall research workflow—covering data acquisition,
256 preprocessing, transformer-based sentiment classification, and multi-layer analytical
257 procedures—is illustrated in Figure 1. In this framework, raw user-generated reviews are
258 transformed into structured and interpretable knowledge through a sequential data-to-knowledge
259 pipeline consisting of preprocessing, transformer-based sentiment representation, model
260 agreement assessment, temporal aggregation, and lexical knowledge extraction. The input layer
261 consists of raw textual reviews, rating scores, and timestamps. The processing layer includes text
262 preprocessing, transformer-based sentiment representation, temporal aggregation, agreement
263 computation, and lexical pattern extraction. The output layer consists of structured knowledge

representations, including temporal sentiment trends, model-contingent classification patterns, agreement–disagreement structures, and lexical indicators of user satisfaction and dissatisfaction.

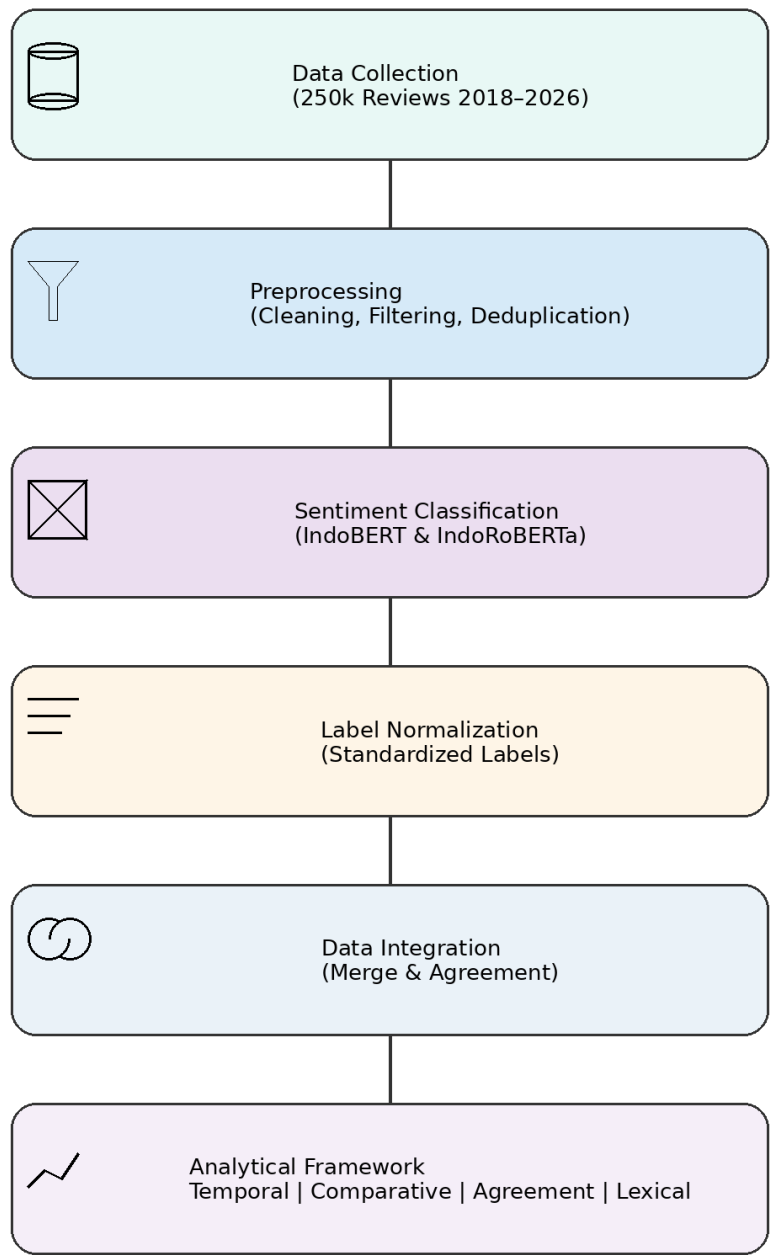


Figure 1. Research Workflow of the Study

3. Results

The results are presented following the proposed multi-layer analytical framework, encompassing temporal, comparative, agreement, and lexical analyses.

3.1 Temporal Dynamics of User Ratings

User evaluation patterns exhibit a clear structural transition over time, characterized by a shift from strong positive dominance in the early period (2018–2020) to increasing negative polarization in later years. As shown in Table 1, the early distribution is strongly skewed toward positive evaluations, reflecting a favorable initial user perception of the platform. The peak of positive sentiment occurs in 2020, where 5-star ratings reach 80.53%, indicating exceptionally high user satisfaction. However, this pattern begins to change in 2021, marking the onset of a structural shift. Although positive ratings remain dominant in that year, the proportion of negative evaluations increases substantially. This transition becomes more pronounced in 2022, where 1-star ratings rise sharply and nearly converge with 5-star ratings, indicating growing dissatisfaction. The most significant reversal occurs during the 2023–2025 period, where negative evaluations become the dominant rating category. In these years, 1-star ratings consistently exceed all other categories, while the proportion of 5-star ratings declines substantially. This pattern reflects a systemic deterioration in user-perceived platform performance rather than a temporary fluctuation.

Table 1. Distribution of Review Scores by Year

Year	Score										Total	
	1		2		3		4		5			
	f	%	f	%	f	%	f	%	f	%	f	%
2018	1035	10.20	482	4.75	705	6.95	1369	13.50	6553	64.60	10144	100.00
2019	8916	22.10	2006	4.97	2681	6.65	4022	9.97	22719	56.31	40344	100.00
2020	9724	10.35	2172	2.31	2640	2.81	3766	4.01	75678	80.53	93980	100.00
2021	10605	22.32	1909	4.02	2345	4.93	3070	6.46	29590	62.27	47519	100.00
2022	8277	39.23	1339	6.35	1343	6.36	1376	6.52	8765	41.54	21100	100.00
2023	5984	58.06	934	9.06	716	6.95	480	4.66	2193	21.28	10307	100.00
2024	8415	60.33	1088	7.80	890	6.38	655	4.70	2901	20.80	13949	100.00
2025	4675	54.29	785	9.12	700	8.13	453	5.26	1998	23.20	8611	100.00
2026	2006	49.58	262	6.48	293	7.24	198	4.89	1287	31.81	4046	100.00
Total	59637	23.85	10977	4.39	12313	4.93	15389	6.16	151684	60.67	250000	100.00

In 2026, a limited recovery is observed, with a slight increase in positive ratings. However, negative evaluations remain relatively high, suggesting that user perception has not fully returned to earlier levels. These temporal patterns are further illustrated in Figure 2.

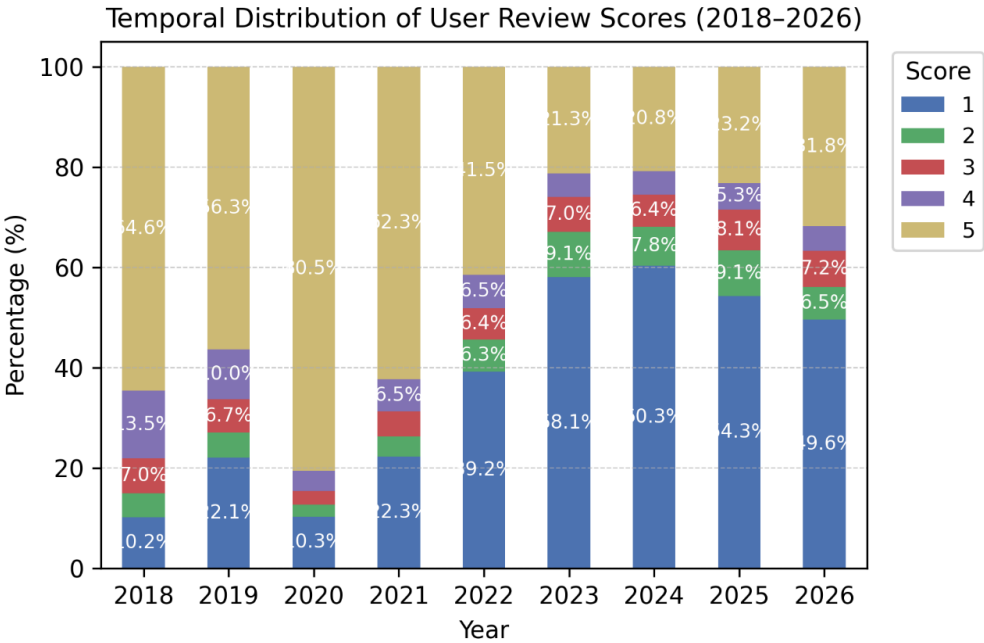


Figure 2. Distribution of User Ratings Over Time

The visualization reinforces the presence of a sustained structural shift, highlighting the progressive decline in positive dominance and the emergence of negative rating concentration in later years. Taken together, these patterns suggest that user evaluation evolves cumulatively over time, shaped by changing platform performance and user expectations.

3.2 Sentiment Classification and Model Comparison

Model-based sentiment classification reveals consistent temporal patterns while simultaneously highlighting systematic differences in how sentiment is interpreted by IndoBERT and IndoRoBERTa. Overall, IndoBERT produces a relatively balanced sentiment distribution, with positive sentiment accounting for 44.90%, followed by neutral (27.68%) and negative (27.42%).

Table 2. Distribution of Sentiment Labels by Year Using IndoBERT (Frequency and Percentage)

Year	Label						Total	
	Negative		Neutral		Positive			
	F	%	F	%	F	%	F	%
2018	2117	20.87	3301	32.54	4726	46.59	10144	100.00
2019	12000	29.74	9164	22.71	19180	47.54	40344	100.00
2020	11360	12.09	30738	32.71	51882	55.21	93980	100.00
2021	11552	24.31	13298	27.98	22669	47.71	47519	100.00
2022	8800	41.71	5371	25.45	6929	32.84	21100	100.00
2023	6101	59.19	2410	23.38	1796	17.43	10307	100.00
2024	8726	62.56	2805	20.11	2418	17.33	13949	100.00
2025	5603	65.07	1474	17.12	1534	17.81	8611	100.00
2026	2285	56.48	643	15.89	1118	27.63	4046	100.00
Total	68544	27.42	69204	27.68	112252	44.90	250000	100.00

In contrast, IndoRoBERTa generates a more polarized distribution, characterized by a higher concentration of both positive (58.47%) and negative (35.93%) sentiment, alongside a significantly reduced neutral category (5.60%).

Table 3. Distribution of Sentiment Labels by Year Using IndoRoBERTa (Frequency and Percentage)

Year	Label						Total	
	Negative		Neutral		Positive			
	F	%	F	%	F	%	F	%
2018	2935	28.93	668	6.59	6541	64.48	10144	100.00
2019	14911	36.96	2184	5.41	23249	57.63	40344	100.00
2020	17264	18.37	6281	6.68	70435	74.95	93980	100.00
2021	15848	33.35	2609	5.49	29062	61.16	47519	100.00
2022	11497	54.49	921	4.36	8682	41.15	21100	100.00
2023	7616	73.89	372	3.61	2319	22.50	10307	100.00
2024	10629	76.20	394	2.82	2926	20.98	13949	100.00
2025	6467	75.10	328	3.81	1816	21.09	8611	100.00
2026	2653	65.57	243	6.01	1150	28.42	4046	100.00
Total	89820	35.93	14000	5.60	146180	58.47	250000	100.00

Across the observation period, both models consistently reflect the same underlying structural shift: positive sentiment dominates in the early period, followed by a marked increase in negative sentiment beginning in 2022. However, IndoRoBERTa consistently amplifies this shift,

producing higher proportions of negative sentiment across all years and exhibiting a more pronounced decline in neutral classifications.

This pattern underscores that model architecture systematically shapes sentiment interpretation rather than merely influencing it. While IndoBERT captures a broader spectrum of sentiment expressions, including intermediate or ambiguous cases, IndoRoBERTa tends to assign sentiment more decisively into polarized categories. As a result, IndoRoBERTa exhibits greater sensitivity to negative signals in user-generated text, particularly in later years.

These findings indicate that sentiment classification is not methodologically neutral, but rather dependent on the interpretative behavior of the underlying model. Consequently, model selection does not merely affect analytical outcomes but fundamentally shapes how user perception is constructed and interpreted. This distinction is particularly important in longitudinal analysis, where model-induced polarization may amplify perceived shifts in sentiment over time. This finding aligns with prior research highlighting the sensitivity of transformer architectures in sentiment representation, while extending it to a longitudinal analytical context.

3.3 Comparative Analysis of Sentiment Distributions

A more detailed comparison between IndoBERT and IndoRoBERTa, as presented in Table 4, reveals systematic and consistent differences in sentiment classification across all years.

Table 4. Comparative Analysis of Sentiment Label Distributions Between IndoBERT and IndoRoBERTa Across Years (Percentage)

Year	Label								
	Negative			Neutral			Positive		
	IndoBERT (%)	IndoRoBERTa (%)	Diff (%)	IndoBERT (%)	IndoRoBERTa (%)	Diff (%)	IndoBERT (%)	IndoRoBERTa (%)	Diff (%)
2018	20.87	28.93	-8.06	32.54	6.59	25.96	46.59	64.48	-17.89
2019	29.74	36.96	-7.22	22.71	5.41	17.30	47.54	57.63	-10.09
2020	12.09	18.37	-6.28	32.71	6.68	26.02	55.21	74.95	-19.74
2021	24.31	33.35	-9.04	27.98	5.49	22.49	47.71	61.16	-13.45
2022	41.71	54.49	-12.78	25.45	4.36	21.09	32.84	41.15	-8.31
2023	59.19	73.89	-14.70	23.38	3.61	19.77	17.43	22.50	-5.07
2024	62.56	76.20	-13.64	20.11	2.82	17.28	17.33	20.98	-3.64
2025	65.07	75.10	-10.03	17.12	3.81	13.31	17.81	21.09	-3.27
2026	56.48	65.57	-9.10	15.89	6.01	9.89	27.63	28.42	-0.79
Total	41.33	51.43	-10.09	24.21	4.98	19.24	34.45	43.59	-9.14

IndoRoBERTa consistently assigns higher proportions of negative sentiment than IndoBERT, with differences reaching double-digit percentages in several years. At the same time, IndoBERT produces substantially higher neutral sentiment classifications, often exceeding IndoRoBERTa by large margins. This pattern indicates a fundamental difference in classification behavior. IndoRoBERTa tends to compress sentiment into extreme categories, while IndoBERT maintains a more distributed representation that includes neutral expressions. The narrowing gap in positive sentiment in later years also points to a gradual convergence in model interpretation as overall sentiment becomes increasingly negative.

These results reinforce the view that sentiment analysis outcomes are shaped not only by data but also by model-specific characteristics. The observed differences highlight the importance of adopting a multi-model approach to obtain a more comprehensive and balanced representation of user sentiment. This divergence is particularly relevant in large-scale sentiment studies, where reliance on a single model may lead to systematically biased interpretations of user perception. To further assess the consistency of these model differences, an agreement analysis is conducted in the following section.

3.4 Agreement Analysis Between Models

Despite the distributional differences identified above, a relatively high level of classification consistency is observed between IndoBERT and IndoRoBERTa.

Table 5. Agreement and Disagreement Between IndoBERT and IndoRoBERTa Classification Results by Year (Frequency and Percentage)

Year	Same		Different		Total	
	F	%	F	%	F	%
2018	7066	69.66	3078	30.34	10144	100.00
2019	31527	78.15	8817	21.85	40344	100.00
2020	67683	72.02	26297	27.98	93980	100.00
2021	35196	74.07	12323	25.93	47519	100.00
2022	15714	74.47	5386	25.53	21100	100.00
2023	7814	75.81	2493	24.19	10307	100.00
2024	11027	79.05	2922	20.95	13949	100.00
2025	7077	82.19	1534	17.81	8611	100.00
2026	3300	81.56	746	18.44	4046	100.00
Total	186404	74.56	63596	25.44	250000	100.00

As shown in Table 5, the overall agreement rate between the two models reaches 74.56%, indicating substantial alignment in sentiment classification. Importantly, the stability of agreement across years indicates that the observed divergence in sentiment distributions (Section 3.3) does not reflect random model inconsistency, but rather systematic differences in how each model interprets sentiment. This suggests that disagreement is structured rather than arbitrary, reinforcing the notion that model-specific characteristics play a critical role in shaping sentiment representation.

Nevertheless, a consistent proportion of disagreement—exceeding 17% in all years—persists throughout the dataset. This indicates that while both models capture similar overarching trends, they differ in how they interpret specific reviews, particularly those containing ambiguous or context-dependent expressions. This finding is particularly important, as it implies that disagreement between models may itself contain analytical value, highlighting cases where sentiment is inherently ambiguous or context-sensitive rather than clearly defined.

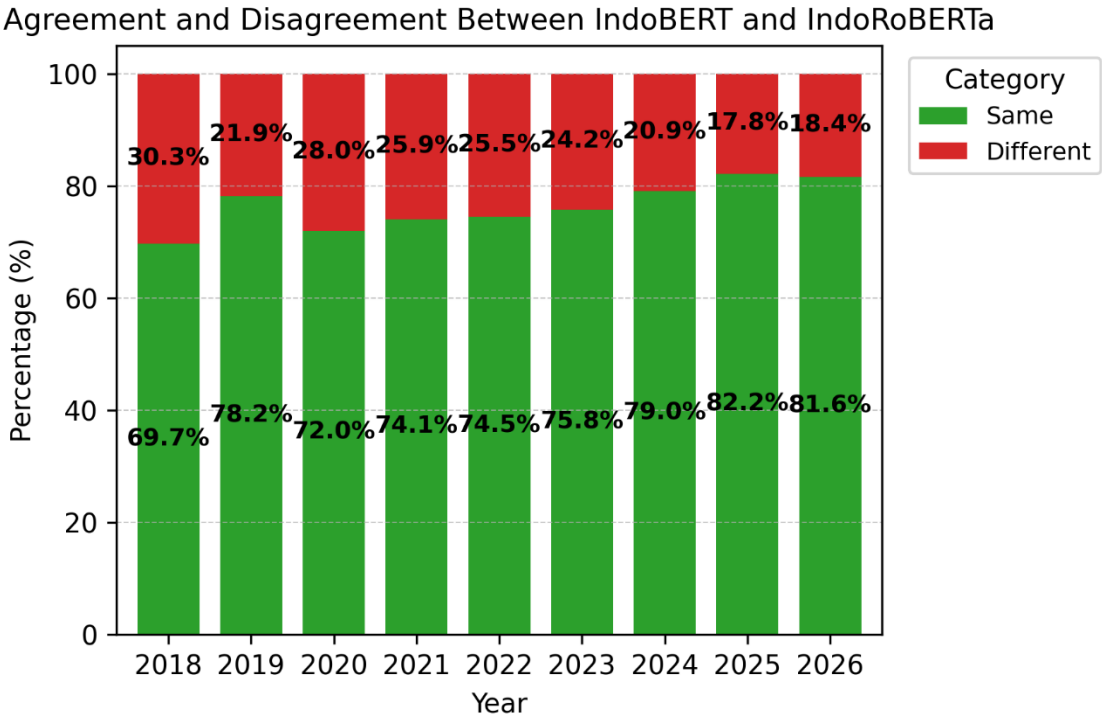


Figure 3. Agreement and Disagreement Between IndoBERT and IndoRoBERTa Classification Results (2018–2026)

These patterns are visually reinforced in Figure 3, which highlights the dominance of agreement alongside the persistent presence of structured disagreement. This pattern suggests that

combining models does not merely improve robustness, but also provides access to complementary interpretative perspectives, particularly in cases where sentiment is inherently ambiguous. Taken together, these findings reinforce the view that sentiment classification should be understood not as a deterministic process, but as an interpretative one shaped by model-specific representations. In this sense, agreement analysis functions as an internal consistency mechanism for assessing the stability of extracted sentiment knowledge across different model representations.

3.5 Lexical Patterns and Behavioral Drivers of Sentiment

Lexical patterns extracted from user reviews provide fine-grained insight into the behavioral drivers underlying sentiment expression, enabling the identification of recurring experiential themes beyond aggregate sentiment classifications.

Positive Sentiment

As illustrated in Figure 4 and detailed in Tables 6 and 7, positive sentiment is strongly associated with expressions of usefulness, satisfaction, and appreciation. Frequently occurring phrases emphasize the perceived value of the platform, particularly in terms of convenience, accessibility, and effectiveness. These patterns indicate that positive sentiment is primarily utility-driven, emerging when the platform effectively delivers functional value to users.

In addition, references to promotional features and value-added services, such as educational programs, suggest that positive sentiment is not solely driven by core platform functionality, but also by supplementary benefits that enhance user experience. This suggests that positive evaluations are structurally anchored in functional performance rather than transient emotional responses, reinforcing the role of utility as a primary determinant of user satisfaction.

Wordcloud of Positive Reviews with Consistent Classification



Figure 4. Wordcloud Representation of Frequently Occurring Terms in Positive User Reviews with Consistent Classification (IndoBERT and IndoRoBERTa)

Table 6. Top 50 Most Frequent Bigrams in Positive User Reviews with Consistent Classification Between IndoBERT and IndoRoBERTa (Frequency and Percentage)

Bigram	Frequency	Percentage (%)
sangat membantu	17668	12.66
terima kasih	8740	6.26
sangat bagus	6896	4.94
toko pedia	6801	4.87
terimakasih tokopedia	6749	4.84
tokopedia sangat	5910	4.24
kasih tokopedia	5224	3.74
sangat bermanfaat	5079	3.64
sangat mudah	4985	3.57
aplikasi sangat	4229	3.03
pelatihan prakerja	3679	2.64
membantu dalam	3051	2.19
belanja tokopedia	3026	2.17
belanja online	2919	2.09
sukses selalu	2816	2.02
skill academy	2710	1.94

sangat memuaskan	2614	1.87
membantu sekali	2612	1.87
sangat baik	2367	1.7
lebih mudah	2097	1.5
mudah dipahami	1875	1.34
sangat puas	1816	1.3
gratis ongkir	1770	1.27
membeli pelatihan	1667	1.19
aplikasi tokopedia	1608	1.15
kartu prakerja	1586	1.14
yg sangat	1569	1.12
tokopedia sudah	1512	1.08
bagus sangat	1463	1.05
bagus banget	1435	1.03
mudah pahami	1417	1.02
banyak promo	1394	1
mengikuti pelatihan	1362	0.98
aplikasi yg	1291	0.93
sukses terus	1253	0.9
bagus sekali	1183	0.85
aplikasi bagus	1172	0.84
bagus mudah	1155	0.83
beli pelatihan	1139	0.82
lebih baik	1124	0.81
pelatihan sangat	1105	0.79
pedia sangat	1092	0.78
mudah cepat	1086	0.78
prakerja sangat	1064	0.76
pelatihan tokopedia	1062	0.76
tokopedia semoga	1048	0.75
barang yg	1045	0.75
sangat senang	1042	0.75
tokopedia memang	1029	0.74
luar biasa	1003	0.72

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Table 7. Top 50 Most Frequent Trigrams in Positive User Reviews with Consistent Classification Between IndoBERT and IndoRoBERTa (Frequency and Percentage)

Trigram	Frequency	Percentage (%)
terima kasih tokopedia	4414	11.86
tokopedia sangat membantu	2633	7.08

sangat membantu dalam	2232	6
sangat membantu sekali	2029	5.45
aplikasi sangat membantu	1737	4.67
aplikasi sangat bagus	1061	2.85
toko pedia sangat	1037	2.79
terimakasih tokopedia sangat	864	2.32
bagus sangat membantu	806	2.17
kasih toko pedia	773	2.08
terimakasih toko pedia	749	2.01
aplikasi yg sangat	698	1.88
terima kasih toko	684	1.84
sangat membantu mudah	666	1.79
sukses selalu tokopedia	641	1.72
sangat bagus sangat	618	1.66
terimakasih tokopedia sudah	583	1.57
belanja toko pedia	569	1.53
pelatihan skill academy	560	1.51
sangat membantu bermanfaat	559	1.5
lebih baik lagi	554	1.49
sangat bagus mudah	549	1.48
tokopedia sangat bagus	547	1.47
kasih tokopedia sangat	532	1.43
sangat bermanfaat sekali	528	1.42
jadi lebih mudah	522	1.4
sangat bagus membantu	509	1.37
membeli pelatihan prakerja	486	1.31
yg sangat membantu	481	1.29
aplikasi tokopedia sangat	473	1.27
tokopedia sangat mudah	465	1.25
tokopedia skill academy	461	1.24
sangat bagus sekali	457	1.23
tokopedia sukses selalu	451	1.21
sangat membantu sangat	448	1.2
sukses terus tokopedia	445	1.2
pedia sangat membantu	436	1.17
belanja lebih mudah	424	1.14
sangat membantu terimakasih	416	1.12
pelatihan prakerja sangat	407	1.09
sangat mudah dipahami	403	1.08
aplikasi belanja online	401	1.08
mengikuti pelatihan prakerja	398	1.07

kasih tokopedia sudah	380	1.02
sudah mengikuti pelatihan	380	1.02
pembelian pelatihan prakerja	364	0.98
terimakasih tokopedia telah	352	0.95
tokopedia sangat bermanfaat	348	0.94
sangat bagus bermanfaat	338	0.91
semoga sukses selalu	337	0.91

These findings indicate that user satisfaction is closely tied to perceived utility and the extent to which the platform delivers meaningful value.

Negative Sentiment

In contrast, negative sentiment is primarily associated with functional issues, unmet expectations, and service inconsistencies, as shown in Figure 5 and Tables 8 and 9. Common expressions reflect recurring problems related to system reliability, feature availability, and delays in service delivery. Notably, phrases indicating deterioration over time suggest that dissatisfaction is cumulative, reflecting ongoing user frustration rather than isolated incidents.

Interestingly, certain features—particularly promotional elements—appear in both positive and negative contexts. This indicates that the same feature can generate contrasting sentiment depending on its consistency and reliability. This duality suggests that user sentiment is not feature-dependent, but performance-dependent, where the consistency of feature delivery determines whether it contributes to satisfaction or dissatisfaction. When expectations are met, such features contribute positively to user experience; when they fail, they become a source of dissatisfaction.

Wordcloud Representation of Frequently Occurring Terms in Negative User Reviews with Consistent Classification



Figure 5. Wordcloud Representation of Frequently Occurring Terms in Negative User Reviews with Consistent Classification (IndoBERT and IndoRoBERTa)

Table 8. Top 50 Most Frequent Bigrams in Negative User Reviews with Consistent Classification (Frequency and Percentage)

Bigram	Frequency	Percentage (%)
tidak bisa	3759	8.28
gratis ongkir	2635	5.81
tidak ada	2546	5.61
pengguna baru	2155	4.75
toko pedia	1809	3.99
gk bisa	1413	3.11
beli barang	1231	2.71
sangat kecewa	1129	2.49
pihak tokopedia	1086	2.39
barang yg	1070	2.36
sama sekali	1058	2.33
tidak sesuai	986	2.17
kurir rekomendasi	952	2.1
lama banget	910	2.01
tolong dong	895	1.97
mau beli	890	1.96
tokopedia care	803	1.77
tolong perbaiki	777	1.71
kasih bintang	763	1.68

belanja tokopedia	760	1.67
mau belanja	747	1.65
lebih baik	744	1.64
yg lain	681	1.5
padahal sudah	678	1.49
barang tidak	667	1.47
aplikasi tokopedia	666	1.47
toko sebelah	634	1.4
ada yg	632	1.39
sangat buruk	628	1.38
setelah update	625	1.38
gk ada	624	1.37
sangat mengecewakan	611	1.35
pas mau	598	1.32
sampai sekarang	586	1.29
oleh sistem	582	1.28
pengembalian dana	577	1.27
sangat tidak	571	1.26
sama tokopedia	567	1.25
tiba tiba	565	1.24
tokopedia tidak	565	1.24
makin kesini	562	1.24
belanja disini	540	1.19
bebas ongkir	535	1.18
bagus tapi	530	1.17
pertama kali	529	1.17
udah bayar	515	1.13
tapi tidak	511	1.13
belanja online	499	1.1
barang sudah	496	1.09
padahal udah	490	1.08

Table 9. Top 50 Most Frequent Trigrams in Negative User Reviews with Consistent Classification (Frequency and Percentage)

Trigram	Frequency	Percentage (%)
makin kesini makin	357	5.34
promo pengguna baru	292	4.36
baru pertama kali	240	3.59
dibatalkan oleh sistem	232	3.47
mau beli barang	191	2.85

tidak bisa digunakan	176	2.63
makin lama makin	173	2.59
kecewa sama tokopedia	172	2.57
ada gratis ongkir	164	2.45
voucher gratis ongkir	163	2.44
tidak bisa dipakai	148	2.21
promo gratis ongkir	148	2.21
pengguna baru tapi	145	2.17
barang tidak sesuai	143	2.14
tidak ada respon	143	2.14
tolong perbaiki lagi	142	2.12
oleh pihak tokopedia	137	2.05
tidak ada solusi	135	2.02
tapi tidak bisa	134	2
tapi tidak ada	131	1.96
tidak memenuhi syarat	128	1.91
setelah update malah	126	1.88
kenapa tidak bisa	126	1.88
barang tidak sampai	125	1.87
buat pengguna baru	125	1.87
tidak bertanggung jawab	120	1.79
tokopedia tidak bisa	120	1.79
gratis ongkir tapi	120	1.79
tidak bisa pakai	117	1.75
sampai sekarang belum	116	1.73
otomatis oleh sistem	111	1.66
kecewa banget sama	110	1.64
aplikasi belanja online	110	1.64
pihak toko pedia	106	1.58
biaya jasa aplikasi	101	1.51
batalan oleh sistem	99	1.48
sama sekali tidak	99	1.48
pertama kali belanja	96	1.43
ada biaya layanan	96	1.43
dibatalkan otomatis oleh	95	1.42
diskon pengguna baru	94	1.4
traktiran pengguna baru	94	1.4
sebagai pengguna baru	94	1.4
tidak bisa gunakan	92	1.37
voucher pengguna baru	88	1.32
jadi males belanja	86	1.29

tolong dong perbaiki	84	1.26
sudah tidak bisa	84	1.26
melanggar syarat ketentuan	82	1.23
aplikasi toko pedia	81	1.21

3.6 Integrated Interpretation of Sentiment Dynamics

When examined collectively, the four analytical layers reveal a coherent structure of sentiment dynamics that cannot be captured through isolated analysis. This integration shows that sentiment dynamics are not produced by a single analytical layer, but emerge from the interaction between temporal change, model-specific interpretation, and recurring linguistic expressions in user reviews.

The findings reveal a consistent longitudinal shift from positive to negative sentiment, supported by both rating distributions and model-based classifications. Importantly, this shift reflects not merely a redistribution of sentiment categories, but a structural transformation in how users evaluate platform performance over time. This shift is accompanied by identifiable behavioral patterns, where dissatisfaction is driven by system-related issues, unmet expectations, and inconsistencies in service delivery, while positive sentiment is linked to perceived usefulness and value-added features.

Moreover, the comparison between IndoBERT and IndoRoBERTa highlights the methodological sensitivity of sentiment analysis, demonstrating that different models produce systematically different representations of user perception. At the same time, the relatively high level of agreement between models suggests that overall sentiment trends remain robust despite these differences. This reinforces that sentiment interpretation is inherently model-contingent, rather than universally objective.

Overall, the results indicate that user perception in digital platforms is shaped by the continuous interaction between functional performance, expectation alignment, and the consistency of platform experience over time. Accordingly, sentiment dynamics should be understood as an evolving feedback system, in which user perception continuously adjusts in response to platform performance, rather than as a static evaluation outcome.

4. Discussion

4.1 Sentiment as a Dynamic and Context-Dependent Construct

A pronounced temporal shift in user sentiment is observed, indicating that user perception evolves dynamically rather than remaining stable. During the early period (2018–2020), positive evaluations were dominant, with 5-star ratings peaking at 80.53% in 2020. This pattern changes

markedly in later years, as negative sentiment rises sharply, reaching 65.07% in 2025 (IndoBERT) and 76.20% in 2024 (IndoRoBERTa). Such a shift reflects a structural transformation in user perception rather than a short-term fluctuation.

This evolution appears closely linked to the developmental trajectory of Tokopedia as a digital platform. In its early growth phase, user evaluations tended to be highly positive, likely due to novelty effects and relatively modest expectations. This reflects a common pattern in digital platform adoption, where initial user enthusiasm is often followed by more critical evaluation as usage experience accumulates (White & Samuel, 2019). As the platform matured and competition intensified, expectations increased accordingly, making inconsistencies in service more visible and more likely to trigger dissatisfaction.

From a theoretical standpoint, this pattern aligns with expectation–confirmation theory, where satisfaction depends on the gap between expected and perceived performance (Ngoc et al., 2023). Dissatisfaction, therefore, emerges not only from service failure but from the increasing gap between evolving user expectations and actual platform performance.

Existing literature further supports this interpretation by showing that online word-of-mouth and user sentiment are shaped by accumulated experience, contextual conditions, and temporal change (Skotis et al., 2023; Stamolampros & Korfiatis, 2018). Rather than remaining fixed, user evaluations tend to evolve as individuals interact repeatedly with digital platforms and adjust their perceptions over time. This perspective reinforces the importance of treating sentiment as a longitudinal construct rather than a static evaluation metric.

4.2 Limitations of Rating-Based Evaluation

A systematic divergence emerges between numerical ratings and textual sentiment classifications, particularly during transitional periods such as 2022–2025. Despite the continued presence of positive ratings, textual evidence points to a clear shift toward negative sentiment.

This discrepancy highlights a fundamental limitation of rating-based evaluation systems. Numerical ratings compress complex user experiences into a single value, often obscuring mixed or context-dependent opinions. By contrast, textual reviews provide richer insight into user perceptions, including specific frustrations and unmet expectations. This limitation becomes particularly critical in longitudinal settings, where gradual shifts in user perception may remain hidden behind seemingly stable rating distributions.

The literature consistently supports this limitation. Chen et al. (2022) demonstrate that online ratings are prone to distributional bias, while Dranove & Jin (2010) show that ratings oversimplify consumer evaluations. António et al. (2018) further confirm that textual reviews offer additional explanatory power beyond numerical scores.

Taken together, the evidence suggests that relying solely on ratings may lead to incomplete or even misleading interpretations of user perception. This is particularly critical in longitudinal contexts, where shifts in sentiment may not be fully captured by numerical indicators alone.

4.3 Multi-Dimensional Nature of User Sentiment

Building on the limitations of rating-based evaluation, a closer examination of textual data reveals that user sentiment is inherently multi-dimensional. The lexical analysis shows that positive sentiment is strongly associated with perceived usefulness and convenience, as reflected in expressions such as “sangat membantu” and “terima kasih.” In contrast, negative sentiment is linked to functional issues, including system errors (“tidak bisa”), unmet expectations (“tidak sesuai”), and delays (“lama banget”). These patterns indicate that sentiment expression is not driven by isolated attributes, but emerges from the interaction between multiple service dimensions that shape overall user experience.

Interestingly, certain features—particularly promotional elements such as “gratis ongkir”—appear in both positive and negative contexts. This indicates that the same feature can generate contrasting sentiment depending on how consistently it meets user expectations. This suggests that service attributes do not carry fixed sentiment values; instead, their evaluative meaning depends on performance consistency and expectation fulfillment.

This interpretation is consistent with prior research highlighting the complexity of sentiment expression (Benamara et al., 2016), while aspect-based sentiment analysis further demonstrates the importance of examining sentiment at a more granular level (Ortiz et al., 2019). More recent studies also emphasize the relevance of aspect-level analysis for capturing the multidimensional nature of user evaluations (Yang et al., 2022; Zhang et al., 2022).

Overall, these findings reinforce the view that user sentiment in digital platforms is multi-layered and driven by interacting service dimensions, rather than isolated factors.

4.4 Model Sensitivity and Methodological Implications

Beyond the sentiment itself, the results highlight that its interpretation is strongly influenced by the analytical model employed. The comparison between IndoBERT and IndoRoBERTa reveals systematic differences in classification outcomes. IndoRoBERTa consistently produces higher proportions of negative sentiment, whereas IndoBERT captures a broader distribution, particularly in the neutral category.

The observed divergence reflects differences in how each model processes contextual information. IndoRoBERTa tends to generate more polarized classifications, while IndoBERT appears more sensitive to intermediate expressions. As a result, model selection serves as an analytical lens rather than a purely technical choice. This suggests that sentiment classification

outcomes should not be interpreted as objective representations of user opinion, but as model-contingent constructions shaped by architectural and training differences.

This observation is supported by prior research demonstrating that transformer-based models capture contextual representations and that their outputs may vary across architectures and training strategies. Studies on Indonesian NLP further indicate that transformer-based models are effective for capturing contextual patterns in Indonesian text (Wicaksono & Rojali, 2026; Yulianti et al., 2024).

Taken together, these findings suggest that sentiment analysis should be understood as an interpretive process shaped by methodological choices, rather than a purely objective analytical procedure. Accordingly, model selection is not merely a technical decision, but a substantive factor that influences how user perception is constructed and interpreted in large-scale textual analysis.

4.5 Comparative Insights with Prior E-Commerce Studies

Situating these findings within the broader e-commerce literature reveals both convergence and extension. Importantly, the present findings show that dissatisfaction is not episodic, but cumulative, reflecting a progressive deterioration in user experience rather than isolated service failures. Consistent with previous studies, user dissatisfaction is closely linked to service reliability, unmet expectations, and platform performance issues (Fan et al., 2013; Netter, 2016). However, these results extend prior work by showing that dissatisfaction accumulates over time rather than occurring as isolated events.

This longitudinal perspective reveals patterns that cannot be captured in cross-sectional analyses, particularly the gradual accumulation of dissatisfaction over time. Studies demonstrate the influence of user-generated content on consumer perception and market outcomes (Akhter & Mohamadali, 2023; Kaur, 2024). However, these studies largely focus on cross-sectional data.

These findings demonstrate that sentiment evolves systematically and reflects cumulative user experience. Accordingly, user perception in digital marketplaces should be interpreted as a cumulative process, where sentiment reflects the aggregation of repeated experiences rather than isolated evaluations.

4.6 Role of Advanced NLP Techniques in Enhancing Insight

Extending from the comparative perspective above, advanced NLP techniques play a crucial role in deepening sentiment analysis. Transformer-based models enable the capture of contextual and semantic nuances that are difficult to identify using traditional methods (Saidi et al., 2025). The key contribution is not merely the application of transformer-based models, but the use of these models to reveal patterns that are unobservable through traditional analytical approaches. In this

study, the use of IndoBERT and IndoRoBERTa enables the identification of temporal shifts, model-sensitive interpretations, and lexical structures that collectively provide a more comprehensive understanding of sentiment dynamics.

The differences observed between IndoBERT and IndoRoBERTa demonstrate that these models do not merely generate classifications, but actively shape how sentiment is interpreted. Their ability to capture linguistic variability makes them particularly suited for analyzing large-scale, real-world textual data (Saidi et al., 2025). This capability is particularly important in such datasets, where user-generated text is highly variable, informal, and context-dependent.

Prior research supports this view (Kyritsis et al., 2023). Transformer-based models demonstrate superiority over traditional methods in sentiment analysis and capture complex semantic patterns (Gupta et al., 2024; Pookduang et al., 2025).

In this context, advanced NLP techniques should be understood not merely as tools for classification but as methodological frameworks that enable the interpretation of complex, large-scale user-generated data in dynamic digital environments.

4.7 Toward a Model-Aware Knowledge Extraction Framework

The findings demonstrate that sentiment dynamics cannot be adequately understood through a single analytical dimension. The framework proposed in this study combines temporal analysis, model comparison, agreement analysis, and lexical exploration into a unified structure. Rather than treating these components as separate procedures, the proposed framework positions them as mutually reinforcing layers of interpretation. From a knowledge engineering perspective, the proposed framework shows that sentiment knowledge is not extracted solely from textual content, but is constructed through the interaction between data characteristics, model representations, temporal aggregation, and lexical structures.

This integrated approach aligns with prior research advocating multi-layer sentiment analysis and emphasizing the importance of combining multiple analytical dimensions (Ashok & Monga, 2022; Williams et al., 2023). By integrating these layers, the framework enables a more robust interpretation of sentiment dynamics by linking when sentiment changes, how models represent these changes, whether interpretations are consistent, and which linguistic patterns explain the underlying behavioral drivers.

4.8 Theoretical and Managerial Implications

The implications of this study extend beyond descriptive findings, offering both theoretical refinement and actionable insights grounded in large-scale empirical evidence.

From a theoretical perspective, the findings reinforce the view that user sentiment should not be conceptualized as a static evaluative construct, but as a dynamic and evolving process shaped by cumulative user experience. The observed longitudinal shift from positive to negative sentiment demonstrates that user perception is continuously reconfigured in response to changes in platform performance and expectation alignment. This extends expectation–confirmation theory by highlighting its temporal dimension, where dissatisfaction emerges not only from isolated service failures, but from the accumulation of unmet expectations over time.

In addition, the results provide methodological implications by demonstrating that sentiment analysis is inherently model-contingent. The systematic differences observed between IndoBERT and IndoRoBERTa confirm that sentiment classification outcomes are shaped by model architecture, rather than representing a single objective reality. This finding supports the need for multi-model analytical approaches, particularly in large-scale and longitudinal studies, to ensure a more robust and balanced interpretation of user perception. Taken together, these implications reposition sentiment analysis from a descriptive tool to a dynamic interpretative framework for understanding user behavior in digital environments.

From a managerial perspective, the findings offer several actionable insights for digital platform providers. First, the consistent association between negative sentiment and system-related issues highlights the critical importance of maintaining platform reliability and minimizing service disruptions. Second, the dual role of promotional features—appearing in both positive and negative contexts—suggests that consistency in feature delivery is more important than feature availability itself. Inconsistent implementation of promotions may erode user trust and amplify dissatisfaction over time.

Third, the results emphasize that monitoring user sentiment should extend beyond numerical ratings to include large-scale analysis of textual feedback. Textual data provide early signals of dissatisfaction that may not be immediately visible in rating distributions, enabling more proactive and data-driven decision-making. Finally, the identification of lexical patterns associated with satisfaction and dissatisfaction suggests that user-generated content can serve as a continuous feedback mechanism for detecting emerging issues and aligning platform strategies with evolving user expectations.

Overall, the findings demonstrate that the ability to systematically analyze and interpret large-scale user-generated data represents not only an analytical capability, but a strategic necessity for sustaining competitiveness in dynamic digital marketplaces.

5. Limitations and Future Research

This study provides a comprehensive analysis of user sentiment dynamics using large-scale review data and advanced transformer-based models. However, several limitations should be

acknowledged. These limitations do not undermine the validity of the findings, but rather define the scope within which the results should be interpreted.

First, this study relies exclusively on user-generated reviews from a single platform, namely Tokopedia. While the dataset is extensive and longitudinal, the findings may not be fully generalizable to other e-commerce platforms or digital ecosystems with different user characteristics, service structures, or market dynamics. Future research could extend the analysis to multiple platforms to enable cross-platform comparisons and enhance generalizability.

Second, the sentiment classification approach is based on pre-trained models, namely IndoBERT and IndoRoBERTa, without domain-specific fine-tuning. Although these models demonstrate strong performance, they may not fully capture platform-specific linguistic nuances, such as informal expressions, sarcasm, or context-dependent meanings commonly found in user reviews. Future studies could improve classification accuracy by incorporating domain-adaptive training or fine-tuning using labeled datasets specific to e-commerce contexts.

Third, this study adopts a single-label sentiment classification framework, categorizing reviews into negative, neutral, and positive sentiments. Such an approach may oversimplify the complexity of user opinions, which often contain mixed or multi-faceted sentiments within a single review. Future research could explore multi-label or aspect-based sentiment analysis to better capture the richness and multidimensional nature of user perception.

Fourth, while this study incorporates temporal analysis, it does not explicitly model causal relationships between sentiment changes and specific platform events, such as policy changes, system updates, or promotional campaigns. Future research could integrate event-based analysis or time-series modeling to better understand the drivers of sentiment fluctuations over time. Taken together, these limitations highlight the trade-off between large-scale exploratory analysis and fine-grained interpretative precision.

Fifth, the lexical analysis conducted in this study focuses on frequent bigrams and trigrams to identify key themes in user reviews. Although this approach provides valuable insights, it may not fully capture deeper semantic structures or latent topics. Future studies could employ more advanced techniques, such as topic modeling, semantic network analysis, or contextual embedding-based clustering, to uncover more nuanced patterns in user feedback.

Finally, while the comparative analysis between IndoBERT and IndoRoBERTa highlights important methodological differences, the study does not evaluate model performance using benchmark metrics such as accuracy, precision, recall, or F1-score due to the absence of labeled ground truth data. Future research could incorporate annotated datasets to conduct a more rigorous evaluation of model performance and validate the robustness of sentiment classification results.

Overall, addressing these limitations would not only enhance methodological robustness, but also enable a deeper and more context-sensitive understanding of sentiment dynamics in digital platforms.

6. Conclusion

This study advances the understanding of user perception dynamics in digital marketplaces by integrating large-scale longitudinal analysis with multi-layer sentiment modeling. Rather than treating sentiment as a static evaluative outcome, the findings demonstrate that user perception evolves cumulatively over time, reflecting the continuous interaction between platform performance, expectation alignment, and user experience.

The results reveal a structural shift from positive to negative sentiment, indicating that dissatisfaction emerges not as an isolated phenomenon, but as a progressive outcome of accumulated user experiences. In addition, the comparative analysis between IndoBERT and IndoRoBERTa demonstrates that sentiment interpretation is inherently model-contingent, highlighting that analytical outcomes are shaped not only by data, but also by the methodological lens through which they are examined.

By integrating temporal, comparative, agreement, and lexical analyses, this study proposes a holistic framework for interpreting large-scale user-generated data. This framework enables a deeper understanding of sentiment dynamics by linking temporal change, model-specific interpretation, and linguistic patterns into a unified analytical structure.

From a practical perspective, the findings underscore the importance of maintaining platform reliability, ensuring consistency in feature delivery, and systematically analyzing textual feedback as an early indicator of user dissatisfaction. These insights highlight that user-generated content is not merely feedback, but a strategic resource for monitoring platform performance and guiding data-driven decision-making.

From a data and knowledge engineering perspective, the proposed framework demonstrates how unstructured user-generated reviews can be systematically transformed into structured knowledge about temporal sentiment dynamics, model-dependent interpretations, and behavioral drivers of user perception.

Taken together, this study demonstrates that sentiment analysis should be understood not simply as a tool for classification, but as a dynamic interpretative approach for capturing the evolving nature of user perception in complex digital environments. In this sense, understanding sentiment dynamics is not only an analytical challenge, but a critical requirement for navigating and sustaining competitiveness in rapidly evolving digital ecosystems.

7. Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used ChatGPT and Jenni AI for language refinement, drafting assistance, and editing support. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the manuscript.

8. Data Availability Statement

The data used in this study were collected from publicly available user reviews on the Google Play Store. Due to platform terms and privacy considerations, the processed dataset is not publicly shared. However, aggregated analytical results and supporting outputs are available from the corresponding author upon reasonable request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.